

Business Requirements Document (BRD)

Project: Pizza Sales Analysis

Project Overview

The Pizza Sales Analysis project is designed to analyse transactional sales data from a pizza store. The goal is to identify key business insights, trends, and KPIs that will help management make informed decisions related to sales, marketing, and operations.

Business Objectives

- 1 Identify overall revenue, total pizzas sold, and total number of orders.
- 2 Determine sales distribution by pizza category, size, and type.
- 3 Analyse time-based trends in sales (daily, monthly, and yearly).
- 4 Highlight best-selling and least-selling pizzas by revenue and quantity.
- 5 Understand customer purchasing behaviour using AOV and average pizzas per order.
- 6 Provide dashboards for effective decision-making.

Data Source & Description

Dataset: pizza_sales.csv

- 1 order_id – Unique identifier for each order
- 2 pizza_id – Unique identifier for each pizza
- 3 pizza_name – Name of the pizza sold
- 4 quantity – Number of pizzas sold per order
- 5 total_price – Total revenue for each transaction
- 6 order_date & order_time – Timestamp for time-based analysis
- 7 pizza_category & pizza_size – Pizza classification attributes

Key Performance Indicators (KPIs)

- 1 Total Revenue = Sum of total_price
- 2 Total Pizzas Sold = Sum of quantity
- 3 Total Orders = Count of unique order_id
- 4 Average Order Value (AOV) = Total Revenue ÷ Total Orders
- 5 Average Pizza per Order = Total Pizzas Sold ÷ Total Orders

Analysis & Visualizations

- 1 Ingredient Analysis – Identify most used ingredients for inventory optimization.
- 2 Daily Trend – Sales trend by day of week for staffing planning.
- 3 Hourly Trend – Sales by hour to identify peak rush times.
- 4 Monthly Trend – Revenue and orders trend to understand seasonality.
- 5 % Sales by Category – Identify dominant pizza categories.
- 6 % Sales by Size – Analyse size-based demand for inventory planning.
- 7 Top 5 Best-Selling Pizzas – Support promotions and menu strategy.

- 8 Bottom 5 Least-Selling Pizzas – Identify improvement or removal opportunities.

Business Questions Answered

- 1 What is the total revenue generated?
- 2 How many pizzas were sold in total?
- 3 Which category and size perform best?
- 4 Which pizzas are top and bottom performers?
- 5 What is the average order value and pizzas per order?
- 6 What are the sales trends by day, month, and hour?

Deliverables

- 1 Jupyter Notebook with complete Python analysis
- 2 Interactive visualizations and charts
- 3 Business Requirements Document (BRD)
- 4 Insights and recommendations for management

Conclusion & Recommendations

The analysis provides a comprehensive view of pizza sales performance. Management can leverage these insights to focus marketing on high-performing categories, optimize the menu, plan inventory and staffing efficiently, and continuously monitor KPIs using dashboards.